

Name.....

Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- MARCH 2024

Eighth Semester

CST 476 MOBILE COMPUTING

Time: 1.5 Hours

Max. Marks: 50

PART A

Answer all questions (3 x 6 = 18 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	Identify the advantages of semantic web.	3	K2	CO1 & K2
2	Distinguish between Infrastructure and ad-hoc wireless networks.	3	K2	CO1 & K2
3	Give any four functions of mobile computing.	3	K1	CO1 & K2
4	Compare and contrast satellite systems-GEO,LEO and MEO.	3	K1	CO2 & K3
5	List the advantages and disadvantages of spread spectrum.	3	K2	CO2 & K3
6	In GSM network, there are some <u>databases</u> used for various purposes. Which are these databases? What are their functions?	3	K3	CO2 & K3

PART B

Answer any two questions (8 x 2 = 16 Marks)

Module 1: Qn. No 7,8 and 9

Qn. No.	Question	Marks	K level of Qn.	CO & KL
7	Draw the three-tier architecture of Mobile computing. Describe the functions of these tiers	8	K2	CO1 & K2
8	What is a middleware? Discuss the role of various types of middleware in mobile computing.	8	K1	CO1 & K2
9	Explain the design considerations for mobile computing	8	K2	CO1 & K2

PART C

Answer any two questions (8 x 2 = 16 Marks)

Module 2: Qn. No 10,11 and 12

Qn. No.	Question	Marks	K level of Qn.	CO & KL
10	Discuss in detail about GSM system architecture	8	K1	CO2 & K3
11	How security is achieved in GSM? Explain	8	K2	CO2 & K3
12	Apply Direct Sequence Spread Spectrum to the data 010 using the Barker Sequence 10011010101. Show the encoding steps.	8	K3	CO2 & K3

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B. TECH DEGREE SECOND INTERNAL TEST- APRIL 2024

Eighth Semester

CST 476 MOBILE COMPUTING

Time: 1.5 Hours

Max. Marks: 50

PART A

Answer all questions (3 x 6 = 18 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	List any three advantages and disadvantages of Wireless LAN.	3	K1	CO3 & K2
2	Compare the features of Flat routing and Hierarchical routing.	3	K1	CO3 & K2
3	Write note on WTLS.	3	K2	CO4 & K2
4	Mention the basic purpose of DHCP. Also list the entities of DHCP.	3	K2	CO4 & K2
5	Explain the basic components of Information security system with a simple diagram.	3	K2	CO5 & K2
6	Give the significance of Snooping TCP with a diagram.	3	K2	CO5 & K2

Part B

Answer any two questions (8 x 2 = 16 Marks)

Module 3: Qn. No 7,8 and 9

Qn. No.	Question	Marks	K level of Qn.	CO & KL
7	Describe the protocol architecture of IEEE 802.11.	8	K2	CO3 & K2
8	Explain about HIPERLAN-1 in detail.	8	K1	CO3 & K2
9	Describe the architecture of Bluetooth.	8	K2	CO4 & K2

PART C

Answer any two questions (8 x 2 = 16 Marks)

Module 4: Qn. No 10,11 and 12

Qn. No.	Question	Marks	K level of Qn.	CO & KL
10	How is IP packet delivery done using Mobile IP?	8	K3	CO4 & K2
11	What are the features of WAP(Wireless Application Protocol). Explain the different layers in WAP architecture with a neat sketch.	8	K2	CO5 & K2
12	How symmetric key cryptography is differ from public key cryptography? Explain any two symmetric key cryptography algorithms.	8	K2	CO5 & K2

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (R,S) / S8 (PT) (R) Examination May 2024 (2019 Scheme)

Course Code: CST476

Course Name: MOBILE COMPUTING

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 ✓ Define mobile computing & list out its advantages. (3)
- 2 ✓ Explain different types of middleware and gateways required in the architecture of mobile computing. (3)
- 3 ✓ Why does the multiple access technologies are important in wireless communication system? (3)
- 4 ✓ Explain the working of Satellite system with diagram. (3)
- 5 ✓ What is meant by routing in Ad-hoc networks? (3)
- 6 ✓ Write a short note on WTLS(Wireless Transport Layer Security). (3)
- 7 ✓ List out the steps involved in DHCP for assigning an IP address to a newly connected host machine. (3)
- 8 ✓ Write about Wireless Datagram Protocol (WDP). (3)
- 9 ✓ What are the major security components of mobile computing? (3)
- 10 ✓ Discuss about the 5G mobile networks. (3)

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Explain any 4 functions and 3 applications of mobile computing. (7)
- b) Describe internet-Ubiquitous networks mentioning the significance and functions of core, edge and access network with diagram. (7)

OR

- 12 a) ✓ Explain in detail the three-tier architecture of mobile computing. (7)
- b) ✓ List out and explain the design issues for mobile computing. (7)

Module II

- 13 a) How does spread spectrum techniques eliminates narrow band interference during signal transmission? (7)
- b) Elaborate the working of following medium access control protocols: (7)
- a. TDMA b. FDMA

OR

- 14 a) Explain about any two Satellite Orbits. (7)
- b) Describe the architecture of GSM network. (7)

Module III

- 15 a) Write the design goals of Wireless LAN and its applications. (7)
- b) Explain the protocol architecture of IEEE 802.11 standard with suitable diagram. (7)

OR

- 16 a) Discuss the architecture of HIPERLAN. (7)
- b) Explain the working procedure and protocol stack architecture of Bluetooth technology with suitable diagrams. (7)

Module IV

- 17 a) Explain the router discovery process in Mobile IP. (7)
- b) Describe the working of DHCP in Mobile computing with neat diagrams (7)

OR

- 18 a) Explain the working procedure of Dynamic Source Routing(DSR) Protocol with suitable example. (7)
- b) How does Wireless Application Protocol(WAP) provide mobility support in mobile applications? Explain with suitable diagram. (7)

Module V

- 19 a) Explain the functioning of Orthogonal frequency division multiplexing. (7)
- b) Explain different security models in mobile computing. (7)

OR

- 20 a) Explain the different security techniques and algorithms used in mobile computing (7)
- b) Describe in detail any 7 pillars of 5G. (7)
